Period/Year	Drone model/system	Developer/country	Technological features	Prima applic
1990s-2000s	*RQ-11 Raven / Civilian Mini-UAVs*	AeroVironment (United States)/Multiple Developers	Electric propulsion, digital control systems, compact sensors, GPS-based autopilot.	Tactica reconn enviro studies mappi

3.3 Principles and Components of Second-generation Drones

The second generation of drones marked a defining shift in UAV design philosophy and functionality, introducing new aerodynamic, electronic, and communication principles that laid the foundation for modern unmanned systems. Unlike the mechanically simple first-generation UAVs, these drones incorporated microelectronics, digital telemetry, and GPS-based navigation, allowing for improved control precision, stability, and operational range. The combination of remote piloting, semi-autonomous flight, and real-time data feedback made this generation a major step toward an intelligent, autonomous aerial platform.

3.3.1 Principles of operation

- *Remote control and data link communication*: The core operating principle of second-generation drones relied on radio frequency (RF) data links for communication between the drone and the ground control station (GCS). Operators transmitted flight commands using multi-channel transmitters, while telemetry data such as altitude, position, and system status was sent back in real time. This two-way